

2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

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[4] ✓ 0s
import pandas as pd
import numpy as np

# Read training data from CSV file
data = pd.read_csv("enjoysport.csv")

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

def learn(concepts, target):
    specific = concepts[0].copy()
    general = [["?" for i in range(len(specific))
               for j in range(len(specific))]

    for i, h in enumerate(concepts):
        if target[i].lower() == "yes":
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    specific[x] = "?"
                    general[x][x] = "?"
        else:
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    general[x][x] = specific[x]
                else:
                    general[x][x] = "?"

    general = [g for g in general if g != ["?" * len(specific)]]

    return specific, general

s_final, g_final = learn(concepts, target)

print("Final Specific Hypothesis:")
print(s_final)

print("\nFinal General Hypothesis:")
print(g_final)
```

OUTPUT:

```
▼ ... Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```